

Industrial Internship Report on

Forecasting of Smart City Traffic Patterns

Domain: Data Science and Machine Learning

Prepared by

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Executive Summary

This report presents the work completed during the Industrial Internship organized by upskill Campus in collaboration with UniConverge Technologies Pvt Ltd (UCT). The assigned project was "**Forecasting of Smart City Traffic Patterns**", which focuses on predicting traffic patterns at multiple city junctions using Machine Learning techniques.

The objective of the project was to analyze historical traffic data, identify traffic patterns, and forecast future traffic volumes to assist government authorities in infrastructure planning and traffic management. The project involved data preprocessing, exploratory data analysis, feature engineering, machine learning model development, performance evaluation, and deployment through an interactive Streamlit dashboard.

A Random Forest Regressor model was implemented and enhanced with holiday-aware forecasting features. The final model achieved excellent prediction performance with an R^2 score of 0.969, demonstrating its effectiveness in forecasting traffic volume. This internship provided practical exposure to data science workflows, machine learning model development, and real-world problem-solving.

TABLE OF CONTENTS

1	Preface.....	3
2	Introduction	4
2.1	About UniConverge Technologies Pvt Ltd	4
2.2	About upskill Campus.....	4
2.3	Objective of the Internship	4
2.4	Reference	4
2.5	Glossary	5
3	Problem Statement	6
4	Existing and Proposed solution	7
5	Proposed Design/ Model	8-9
6	Performance Test	10
7	My learnings	11
8	Outputs	12-17
9.	Future Work Scope.....	18
10.	Conclusion.....	19

1 Preface

The Industrial Internship program provided an opportunity to gain practical experience in solving real-world industry problems using modern technologies. During the internship, I worked on a **Forecasting of Smart City Traffic Patterns** project aimed at improving urban traffic management through data-driven decision making.

The project required understanding traffic patterns across multiple city junctions and building a predictive system capable of forecasting future traffic volumes. Throughout the internship, I learned data preprocessing, exploratory data analysis, feature engineering, machine learning model development, evaluation techniques, and dashboard deployment.

This internship significantly enhanced my technical and analytical skills while providing valuable exposure to industry-standard tools and methodologies.

I would like to express my sincere gratitude to upskill Campus, UniConverge Technologies Pvt Ltd (UCT), my mentors, and faculty members for their guidance and support throughout the internship.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT) is a technology company specializing in Digital Transformation, Industrial IoT, Cloud Computing, Machine Learning, and Smart City solutions. The company provides innovative technology solutions for industries and government organizations to improve efficiency, sustainability, and decision-making.

2.2 About upskill Campus

upskill Campus is a career development platform that offers internships, training programs, and industry-oriented learning opportunities. It helps students gain practical experience through real-world projects and mentorship from industry experts.

2.3 Objective of the Internship

The objectives of this internship were:

- To gain practical experience in Data Science and Machine Learning.
- To solve a real-world smart city problem.
- To understand the complete machine learning project lifecycle.
- To improve analytical, programming, and problem-solving skills.
- To develop a deployable predictive analytics solution.

2.4 References

1. Kaggle Smart City Traffic Patterns Dataset.
2. Python Documentation.
3. Scikit-Learn Documentation.
4. Pandas Documentation.
5. Streamlit Documentation.

2.5 Glossary

Term	Description
EDA	Exploratory Data Analysis
ML	Machine Learning
RF	Random Forest
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
R^2	Coefficient of Determination

3. Problem Statement

The government aims to transform cities into smart cities by utilizing digital technologies to improve public services and infrastructure. One of the major challenges faced by urban areas is traffic congestion.

The objective of this project is to forecast traffic volume at four city junctions using historical traffic data. Understanding traffic patterns helps authorities:

- Predict traffic peaks.
- Optimize traffic signal timing.
- Improve road infrastructure planning.
- Reduce congestion.
- Enhance citizen mobility.

Traffic patterns vary significantly based on weekdays, weekends, holidays, and seasonal factors. Therefore, accurate forecasting is essential for effective traffic management.

4. Existing and Proposed Solution

Existing Solution

Traditional traffic management systems rely on historical observations and manual analysis. Such approaches often fail to accurately predict future traffic conditions.

Limitations

- Reactive rather than proactive management.
- Inability to forecast future traffic volumes.
- Limited consideration of temporal patterns.
- Poor scalability.

Proposed Solution

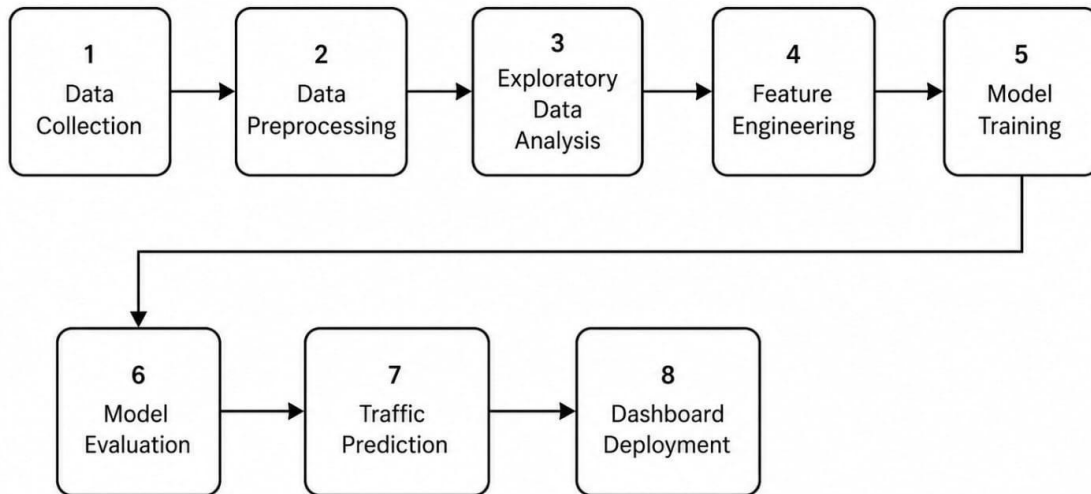
The proposed solution utilizes Machine Learning techniques to forecast traffic volumes using historical traffic data.

Key Features

- Automated traffic prediction.
- Holiday-aware forecasting.
- Traffic trend analysis.
- Interactive dashboard visualization.
- Support for infrastructure planning.

5. Proposed Design / Model

System Workflow



Feature Engineering

The following features were extracted from DateTime:

- Year
- Month
- Day
- Hour
- Day of Week
- Weekend Indicator
- Holiday Indicator
- Junction Number

Machine Learning Model

Random Forest Regressor

Random Forest was selected because:

- Handles non-linear relationships effectively.
- Works well with tabular datasets.
- Reduces overfitting.
- Provides feature importance analysis.

6. Performance Test

Test Plan

The dataset was divided into training and testing subsets. The Random Forest model was trained using engineered features and evaluated using standard regression metrics.

Test Procedure

1. Load and preprocess data.
2. Extract time-based features.
3. Add holiday indicators.
4. Split data into training and testing sets.
5. Train Random Forest model.
6. Evaluate model performance.

Performance Outcome

Metric	Value
MAE	2.39
RMSE	3.54
R ² Score	0.969

Interpretation

The model achieved excellent performance with an R² score of 96.9%, indicating that it accurately captures traffic patterns and provides reliable forecasts.

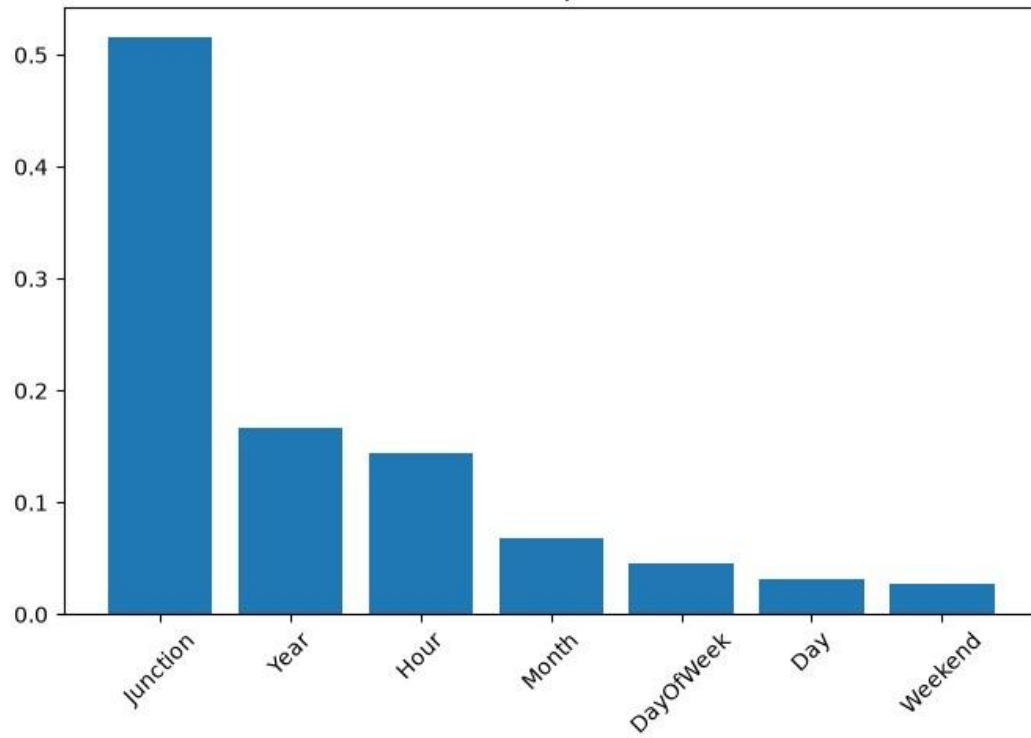
7. My Learnings

Throughout this internship, I gained practical experience in:

- Data Cleaning and Preprocessing
- Feature Engineering
- Exploratory Data Analysis
- Machine Learning Model Development
- Model Evaluation
- Random Forest Algorithm
- Holiday-Aware Forecasting
- Streamlit Dashboard Development
- Real-World Problem Solving

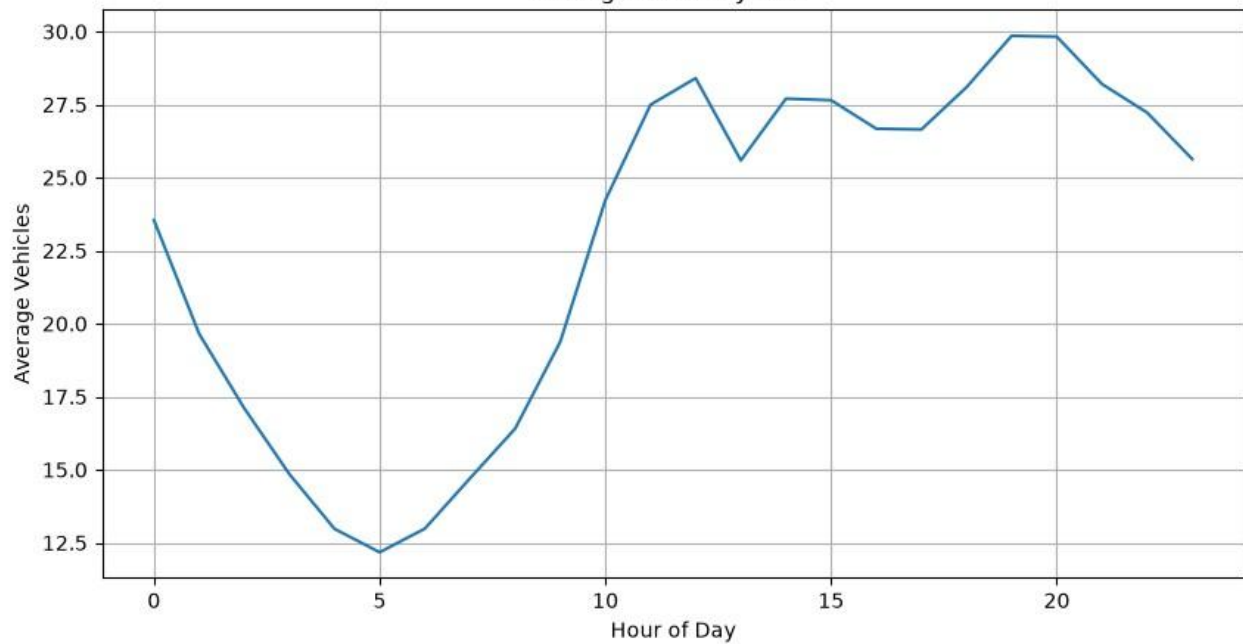
The project improved my understanding of the complete data science lifecycle and enhanced my technical confidence.

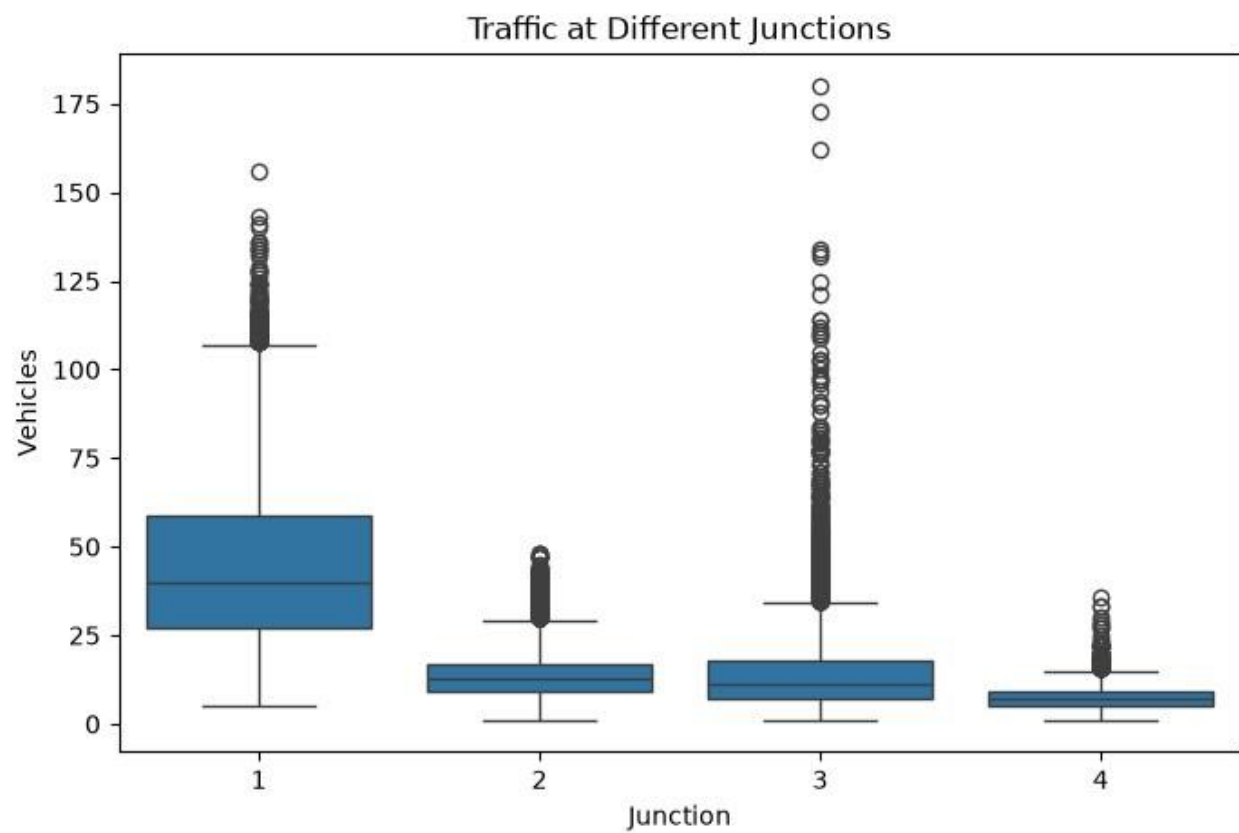
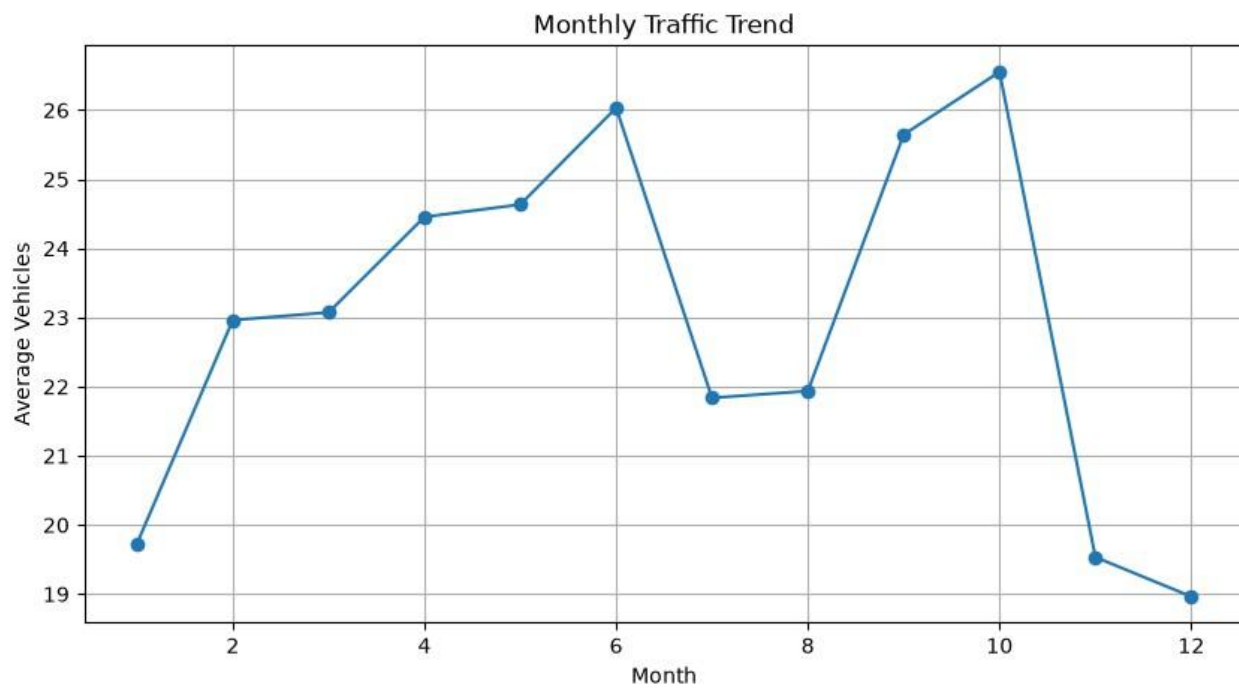
Feature Importance



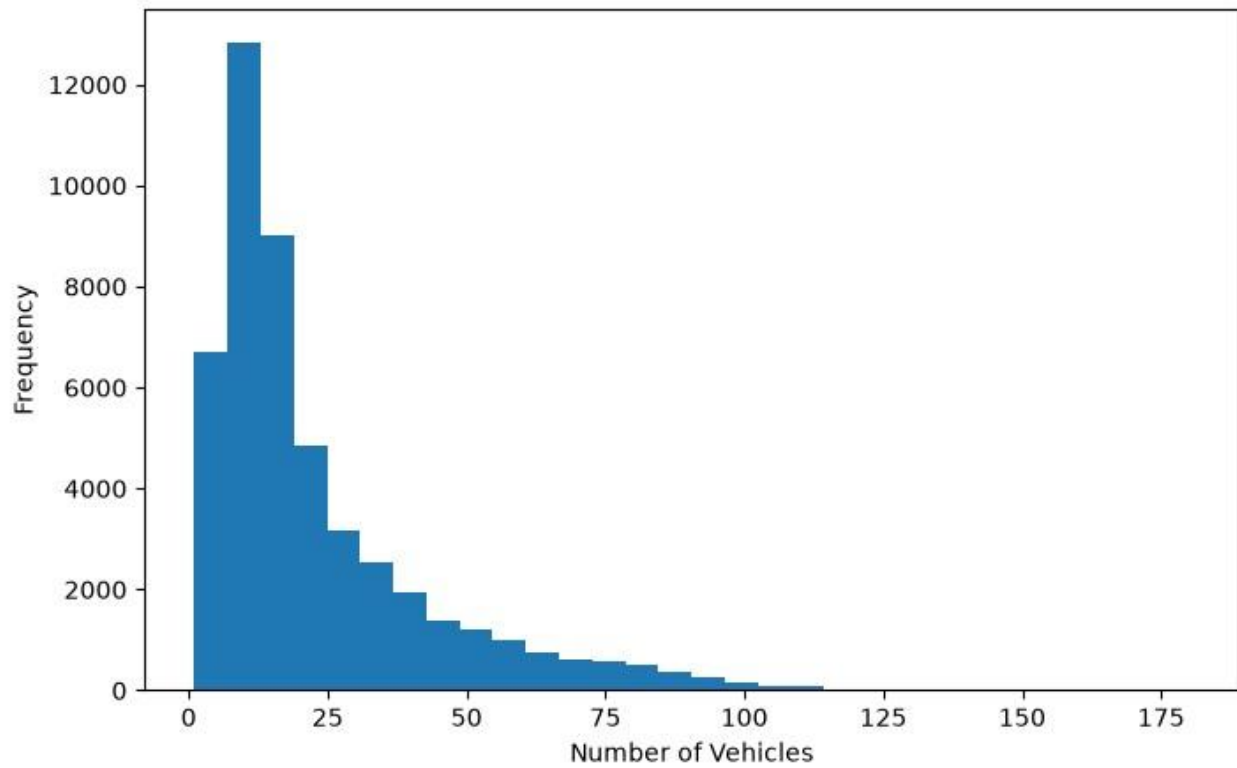
Das

Average Traffic by Hour

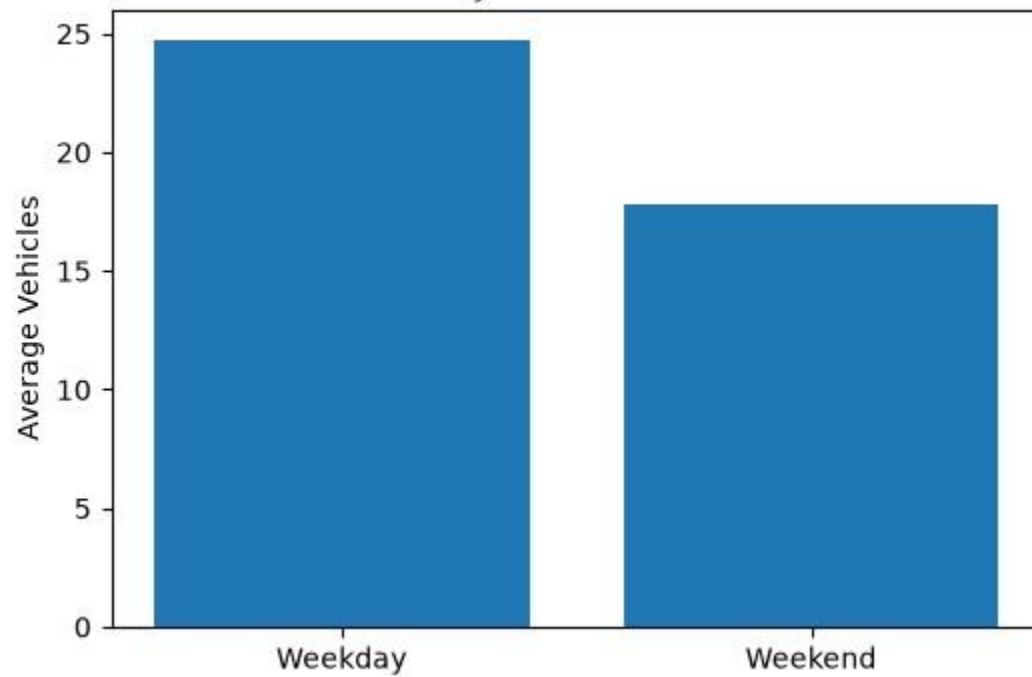




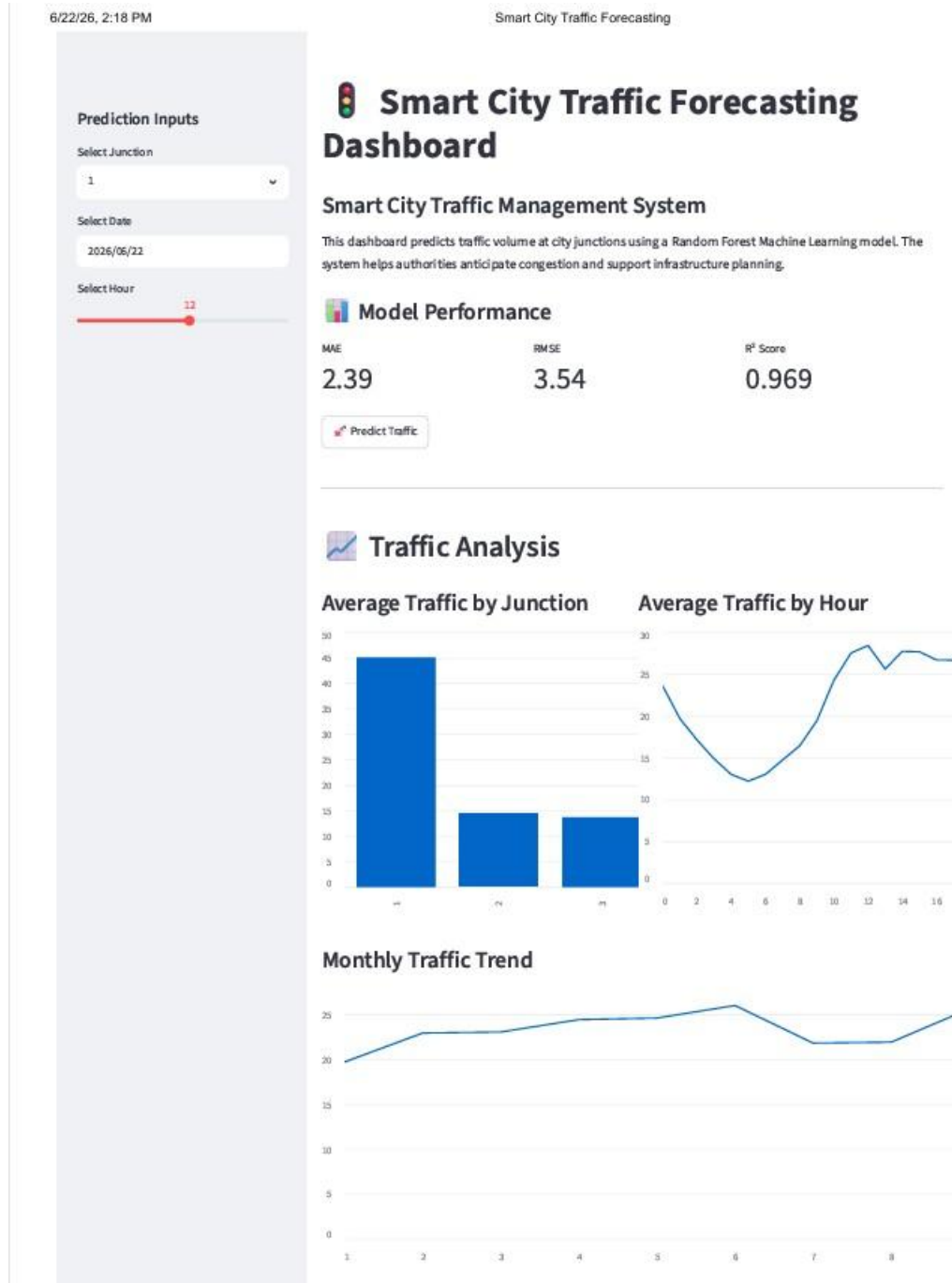
Traffic Distribution



Weekday vs Weekend Traffic



Dashboard:



6/22/26, 2:18 PM

Smart City Traffic Forecasting

Weekend vs Weekday Traffic



Project Information

Project Title: Forecasting of Smart City Traffic Patterns

Objective: Forecast traffic patterns across four city junctions to help city authorities manage congestion and improve infrastructure planning.

Machine Learning Model: Random Forest Regressor

Features Used:

- Junction
- Year
- Month
- Day
- Hour
- Day Of Week
- Weekend Indicator
- Holiday Indicator (IsHoliday)

Key Outcomes:

- Identified traffic peaks.
- Analyzed traffic trends across four junctions.
- Incorporated holiday effects.
- Generated future traffic forecasts.
- Built an interactive dashboard for decision-making.

✓ Internship Project Completed Successfully

9. Future Work Scope

Several improvements can further enhance the project:

- Integration of Real-Time Traffic Data
- Weather-Based Traffic Forecasting
- Event-Based Traffic Prediction
- Deep Learning Models (LSTM)
- SARIMA Time Series Forecasting
- Cloud Deployment
- Mobile Application Integration
- Smart Route Recommendation System

These enhancements would make the system more suitable for large-scale smart city implementations.

10. Conclusion

The Smart City Traffic Forecasting project successfully demonstrated the application of Machine Learning in solving real-world urban traffic challenges. By leveraging historical traffic data, feature engineering, holiday information, and Random Forest Regression, an accurate forecasting model was developed.

The final solution achieved high prediction accuracy and was deployed through an interactive Streamlit dashboard, making it useful for city planners and decision-makers. This internship provided valuable industry exposure and strengthened my skills in Data Science, Machine Learning, and Software Development.

Run Commands: open vs code terminal type **streamlit run app.py**

Repository Name: upskillcampus

Git Link: <https://github.com/Supriya-2604/upskillcampus.git>